—and was all along—an element of reality that says “spin-right for Bob’s qubit if it’s measured along the horizontal axis.”

There’s no way for either Alice’s particle or Bob’s to know ahead of time which measurement Alice was going to make. Hence, Bob’s qubit must come equipped with elements of reality guaranteeing that its spin would be up if measured vertically, and right if measured horizontally.

That’s exactly what the uncertainty principle says cannot happen. If the vertical spin is exactly determined, the horizontal spin is completely unknown, and vice versa, at least according to the conventional rules of quantum mechanics. There is nothing in the quantum formalism that can determine both a vertical spin and a horizontal spin at the same time. Therefore, EPR triumphantly conclude, there must be something missing—quantum mechanics cannot be a complete description of physical reality.

The EPR paper caused a stir that reached far beyond the community of professional physicists. The *New York Times*, having been tipped off by Podolsky, published a front-page story about the ideas. This outraged Einstein, who penned a stern letter that the *Times* published, in which he decried advance discussion of scientific results in the “secular press.” It’s been said that he never spoke to Podolsky again.



EINSTEIN ATTACKS
QUANTUM THEORY

Scientist and Two Colleagues
Find It Is Not ‘Complete’
Even Though ‘Correct.’

SEE FULLER ONE POSSIBLE

Believe a Whole Description of
‘the Physical Reality’ Can Be
Provided Eventually.

(Courtesy of Wikipedia)